

Air to Water Extraction in Abalessa, Algeria

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Problem Statement:

The primary problem we are trying to address is the lack of clean, sustainable, accessible drinking water for those in southern Algeria. The area we will specifically be focusing on is the Town of Abalessa Algeria that is within the Tamanrasset Province. This town contains a population of about 9000 people and is rated by ThinkHazard as an area where water scarcity is considered a high level hazard. As constituted by the website, a high hazard level means drought can be expected every five years and all “project design, and construction methods must take into account the level of water scarcity” as it plays that large of a role in the functioning of the town. However, this is only the start of the problem as “climate change, human stressors and agricultural activities” are making water scarcity and quality an increasingly prominent problem (Zettam, A. et al. (2022)). Agricultural activities are heavily prominent in these small rural towns as they are a primary method of income for many in the town. But, many fertilizers that are necessary to grow crops in these arid conditions can pollute the local watershed by turning the water from “oligotrophic to eutrophic” which poses innate health risks from added nitrogen. These risks can be associated with a nitrate level exceeding “11.3 mg l-1 (equivalent to 50 mg nitrate l-1)” which is the limit set by the World Health Organization and the EU Nitrate Directive (Jenkinson, D. S. (2001)). It has been shown that an increase in Nitrate in one's diet can exhibit a “prophylactic effect against gastrointestinal pathogens” and has been suspected to promote “stomach or esophageal cancers” though that has not been explicitly proven as it is still being researched (Jenkinson, D. S. (2001)). On top of nitrates, there are other pollutants that are found in high concentrations in the sediment of large areas of Algeria such as zinc, cadmium and lead (Boudebouz, A (2024)). These chemicals can provide a wide range of negative health effects and are present in high concentrations for various reasons. Zinc is largely in the local sediment and water because of improper waste disposal which is touched on more in the following paragraph. Cadmium has been shown to cause “testicular tumors, renal dysfunction, hypertension, arteriosclerosis, growth inhibition, chronic diseases of old age, and cancer” when absorbed into the body. Finally lead in drinking water can provide neurodevelopmental health risk in many children as showcased by the fact that “children with a blood lead level greater than 2 µg/dL (0.10 µmol/L) were at a 4.1-fold increased risk of attention-deficit hyperactivity disorder” and that “intelligence quotient point decrement of 3.9 (95% CI 2.4–5.3) associated with an increase in blood lead levels from 2.4 to 10 µg/dL (0.12 to 0.48 µmol/L)” (Payne, M. (2008)).

There are also a lot of other pollution effects that can be attributed to Algeria's rapid industrialization and growth. One of the largest of these effects is improper management of solid waste. Waste dump contaminates are some of the most dangerous for their toxicity and non-degradable nature (Sankhla (2019)). This means they are stored in the bodies of many species lower on the food chain and are passed on in large concentrations to higher predators through a process called biomagnification. This is how much of the aforementioned zinc contaminant gets into human bodies as we are the dominant predator. This means these contaminants are intaken by not only drinking this unfiltered water, but eating the animals that do the same. In the more industrial northern Algeria this release of solid waste is immense as “in

Algiers Bay alone, only 8 % of wastewater undergoes treatment before disposal” which means in a less developed environment the situation is likely even worse because of a lack of water treatment resources at all (Boudebouz, A (2024)).

While all of these contaminants are a large reason for the poor water quality, the quantity of the water is also a problem. Unsustainable removal of water from the natural watershed will disrupt the water cycle of the area and eventually result in depletion of natural water reserves. This is because currently Algerians are taking 38% more water than the natural refresh rate for the area (UN Water (2022)). Not only is the ground water being unsustainably overused there is also going to be less water in the coming years due to increasing climate change. Through the data collected from a study on Science Direct, experts “predict that rainfall events are less frequent but more intense, while droughts are more common and longer” which will cause and or worsen societal problems such as, “water resources, declining agricultural yields” and the “encroaching desert” (Sahnoune, F (2013)).

The agricultural aspect will also cause a chain of economic problems in terms of imports and a lack of exports. Algeria builds most of its money from oil and gas sales yet when “the top imports of Algeria are Wheat (\$1.83B), Cars (\$1.58B), Concentrated Milk (\$1.31B), Corn (\$1.07B), and Raw Sugar (\$932M)” It can be noted that it spends much of this money on agricultural products (Algeria (DZA) Exports, Imports, and Trade Partners (2022)). The agricultural market in Algeria has stagnated so severely that experts are saying that “several warning signs show that the sector is only surviving thanks to massive imports of milk, seeds, grains and chemical fertilizers” (BOUCHENTOUF, S (2021)). This can be seen more precisely when looking at the switch of the country from a mass exporter to a mass importer between 2000 and 2020. In 2000, Algeria bought 9 billion dollars in imports and sold 22 billion in exports, whereas in 2020 they had spent 34 billion on imports and sold 24 billion on exports (View of Analyzing... (2020)). This correlates with the steady decrease in available m3 of water per resident per year which was reported as to be 500 in 2020 and 430 in 2013 (BOUCHENTOUF, S., & BENABDELI, K. (2021)). Because most of the country's imports are agricultural, and the massive boom of imports each year is resulting in significant deficit spending of the country I believe the economic negatives purely related to water scarcity constitute a significantly large problem that, when combined with the issues of water quality and pollution, is worth solving.

Dust storms are a somewhat common danger for Algerians. In the spring, they happen at least once a week, with decreased frequency during the rest of the year (Goudie, A. S., & Middleton, N. J. (2001)). They are very dangerous for people finding themselves outside during the storm, as exposure to the dust can cause anything from coughing to lung diseases (Association, A. L. (n.d.)). The dust is problematic for a while as well, since the particles from the storm can stay in the air for up to a week (Association, A. L. (n.d.)). Furthermore, the dust can accumulate in buckets of water, contaminating the water. This is problematic, since the main source of water currently is groundwater, which people must acquire through a well and bring back to their house, making that water more easily contaminated. Also, with the particles staying in the air for up to a week, even the water they gather after the storm can get contaminated with the dust. The American Lung Association says to seek shelter and if you must, drive carefully,

as in addition to the lung irritation, the dust can drastically reduce visibility, making walking or driving potentially dangerous. If someone is out of water, they have to walk to get it, directly against the directions of the ALA, meaning that they have to choose between having water to cook and drink or being safe and staying out of the storm.

Approach:

Technology breakdown and Energy balance:

This technology uses a certain material called zeolite which is a composite material made of aluminum, phosphorus and iron. Zeolite is incredibly adsorbent which means a lot of water collects on its surface. This is unlike an absorbent material in which the water would collect inside of its volume. This design uses this material along with the natural temperature fluctuations of the day and night to save energy and optimize usage. During the night lots of air is blown through the layers of the zeolite and it slowly collects water from the air passing by on its surface. Then, during the day the device closes itself so that no air can get in or out. Next, the top solar absorber plate takes the heat from the sun to slowly warm the top of the now wet zeolite plates. Because the water is stored solely on the surface of the material, it's much easier for the water to come off. This means that, when heated, the box becomes saturated with a very wet vapor which then gets collected on the cold condensing plate on the top of the second layer of zeolite. This lower layer then heats up due to the heat released from the condensation of water. This causes the condensed water attached to its surface to also turn into vapor and condense on the copper plate below it. This continues on all layers and is the largest reason why this design balances energy very well. It doesn't heat and cool each layer individually, and instead uses a chain reaction of heating to condensation and condensation to heating. It also uses the temperature fluctuations of the cycle of a full day to regulate the temperature it needs. So because of these two facts it massively optimizes its energy balance and makes the machine able to run off nothing but the sun. After all the condensation on each plate is complete, it is funneled into the desired water storage medium. This device works well until humidity of 20% or less so it will still be difficult to run in Abalessa Algeria which currently sits at 9% humidity. Because we have no way of quantifying how much the efficiency will be reduced at an untested humidity, we will assume the same 0.8 liters per square meter that were concluded at the undisclosed temperature and humidity in the MIT area when these tests were conducted.

Amount of water needed per day = 640 L

Latent heat of water = 2260 kJ/L

$Q = mL_v$ (energy needed = mass of substance x latent heat of material)

(Latent Heat of Water: Definition, Formula, Forms of Latent Heat. (2025))

Maximum Energy needed = $2260 * 640 = 1446400$ kJ of energy per day

800 M² of zeolite total

5 layers per machine

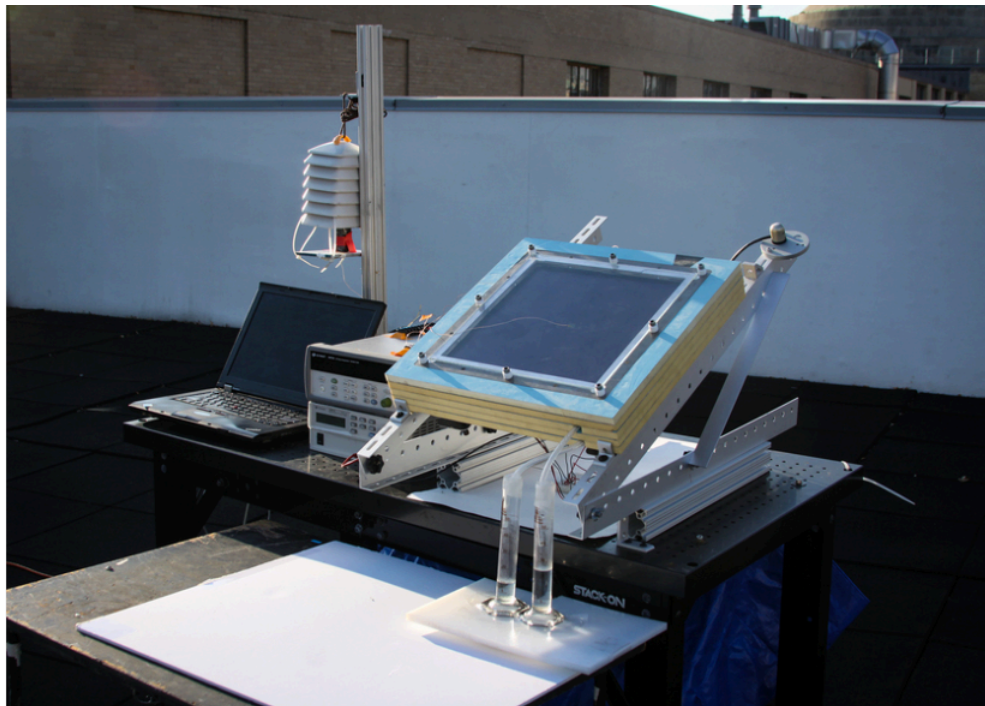
$800/5 = 160$ M² of solar absorbent top layer

$$1446400/160 = 9040 \text{ J/M}^2$$

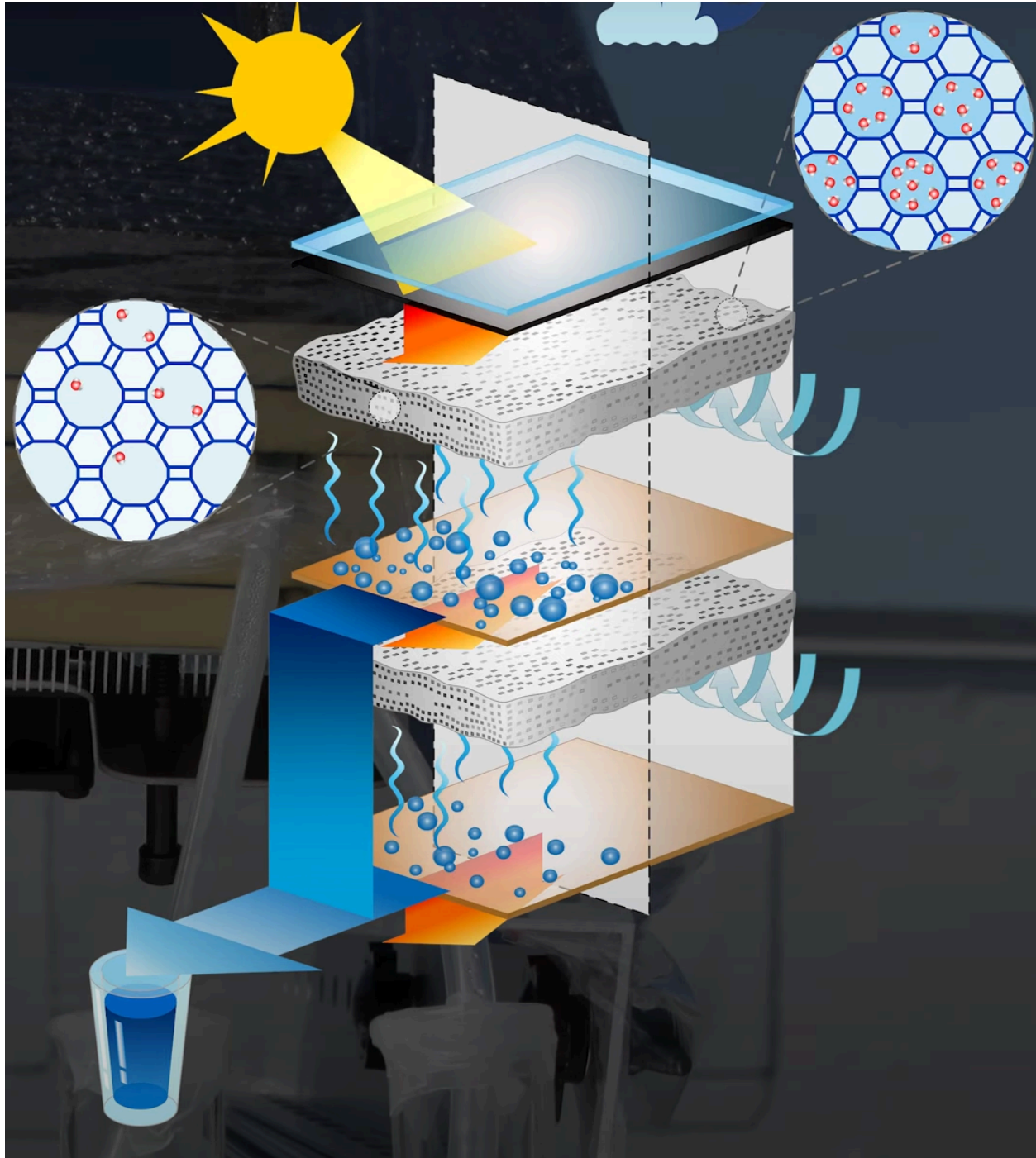
According to the literature about the technology, all of this energy is provided from the sun during the day. According to WorldData, Algeria gets around 13h per day of sunlight throughout the year which means, in total, the sun needs to supply 111261.5 kJ of energy per hour or 30905 Watts in order to condense all of this water.

Economic Cost of Operation:

In order to evaluate the full cost of various layers of the operation we first need to outline the full scale of the project. The total population of the town of Abalessa of Algeria has around 9000 people which is far greater than the amount that this method will be able to serve. So, we chose to isolate the size down to a smaller subsection of around 100 people. According to the mayo clinic the average amount of water the average person needs to drink is around 3.2 liters. Obviously we are not only providing water for drinking, but as this water is incredibly pure, it would probably be best, at the start, to use it as purely food and drink related. For this, we will try to provide around 6.4 liters of water per person per day. This means 640 L per day and if the machines make around 0.8L of water per square meter per day, we need 800 square meters of zeolite to accommodate this group of people. According to listings by The Lab Depot zeolite sheets cost about \$140 per 200mm by 200mm square (PA6.6 Membrane Sheet, 200x200mm. (2025). This means that the cost of the zeolite alone would be $800/(0.2)^2 * 140 = 2.8$ million dollars. If we follow the design given by MIT in the following images, we would also need 800 m² of thin copper plating and a perimeter of likely half inch thick wood for every layer in all the machines.



(Chandler, D. (2020))



(Chandler, D. (2020))

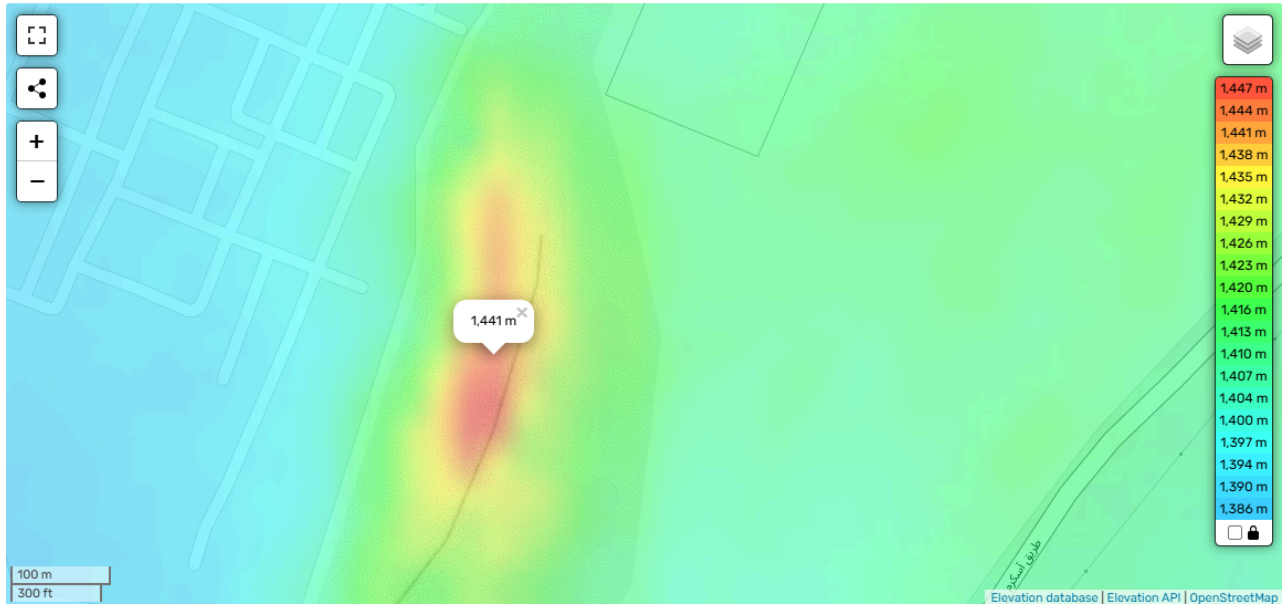
If we use copper foil as our copper plating we can get about 0.2 m^2 for around 19\$ according to a listing on amazon (Amazon.com: 99.99% Pure Copper Foil for Crafts, 0.003"x4"x79"...(2025)) We would need at least 400 of these rolls which is around 7.6 thousand dollars. In order to calculate the cost for the wood we need to know the dimensions of each of the ten machines we intend to make. If we want to have the smallest perimeter possible, to

minimize cost, we should make them square. That means, if each needs to have a total area of 80 m^2 and they have 5 layers, each layer will need to be $4 \text{ m} \times 4 \text{ m}$. Assuming a perimeter thickness of 4 cm we will need 0.634 m^2 of wood per layer per machine. So, with 5 layers per machine and 10 machines, we need a total of 31 m^2 of wood. If we use around a $\frac{3}{4}$ inch thickness for the wood sheets, we can get about 0.47 m^2 of pressure treated (water resistant) wood for 24 dollars according to a listing on home depot ($\frac{3}{4} \text{ in.} \times 2 \text{ ft.} \times 4 \text{ ft.}$ Pressure-Treated Plywood Project Panel. (2022)). We need about 68 of these sheets which comes out to 1.6 thousand dollars.

In total this brings the total cost of all the machines to approximately \$2,809,200. However, It has been found that Algeria is rich in a specific type of clay which can be used to synthesize zeolite through the usage of hydrothermal treatment. This is great because synthetic zeolites are used much more than other, more natural, forms in commercial means. Because it has access to this clay Algeria can have a solid foot in the synthesized zeolite market with help from other more developed countries to aid in the chemical processes needed (Dali Youcef, 2020). Because of this they can likely acquire zeolite easier than many other areas on the globe and as that is the primary material needed in the water condensing machines discussed above, I believe this is one of the best options they have access to, especially when combined with the energy efficiency of this device.

Transportation from zeolite collection machines to storage tank and placement of zeolite machines:

Most of the population lives on one of the flattest parts in the full region so attaching a large set of these devices in one area connected to a central water storage unit would likely be difficult as it would either need to be located on far off mountains or pumped up a high tower which would take a lot of energy and time. However there is a sizable hill nearby that is about 50 meters above the rest of the town and is probably our best course of action if we make the sample size the neighborhood closest to the hill. This hill has a lot of area on its top face as can be seen by the following image.



(topographic-map.com. (2025))

We can see by the scale of the map that this hill is around 75 meters by 2-3 hundred meters. This adds up to around 15-20 thousand square meters of area.

Given that 800 square meters of zeolite are needed to collect the water required for the 100 people and that the machine normally uses 5 layers of the zeolite, 160 square meters of the machine will be required, or about a 13 meter square. To save on energy costs, the machines should be placed above the water tower on the hill, so gravity can be used to move the water to the tank, negating the need for a pump. 640L per day is needed and the machine only allows water to be deposited during the day, so the pipes must carry all 640 L in half of the available time, about 12 hours a day.

$$\frac{640L}{12h} = 53.3 L/h$$

$$\frac{53.3}{60} = 0.88L/min$$

The average house elevation is about 1392 m, and the hill's max elevation is about 1446 m. As such, the top of the hill is about 54 m above the houses. Since the tower is supposed to be 38 meters above the houses and the hill next to the area the tower services is about 54 meters above the houses, the tower can be placed on the hill 38 meters up, allowing the zeolite to be built on top of the hill 16 meters above the tank. The top of the hill is about 40 m by 80 m or 3200 m², more than enough for the 160 m² needed for the zeolite devices.

$$50 \text{ m across, } 16 \text{ m down, length} = 52.49 \text{ m} \rightarrow \sqrt{50^2 + 16^2}$$

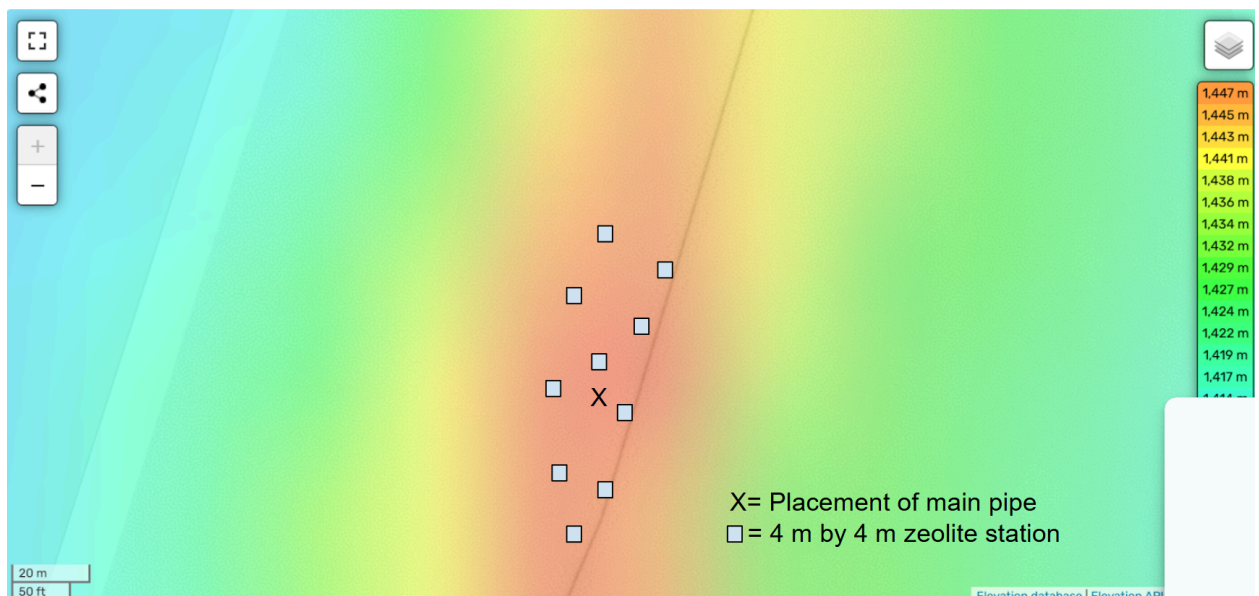
$$\text{Reynold's Number: } Dvp = 0.06 * 3.847 * 10000.009204 = 267700$$

$$\text{Surface roughness} = 0.0015 \text{ mm}$$

Friction factor using moody diagram = 0.0144

$$\text{Energy loss due to friction} = f * \frac{L}{D} * \frac{v^2}{2g} = (0.0144) * \frac{52.49}{3847} * \frac{.06^2}{2(9.8)} = 9.51m < 16m$$

Since the energy loss due to friction is less than the height difference, gravity alone is enough to get the water from the zeolite machines to the tank. Were the energy loss higher than the height difference, a pump would be required to force the water into the tank, increasing the energy cost drastically. Using a 6 cm diameter PVC pipe will allow the water from the zeolite to flow downhill to the tank using only gravity, negating the need for a pump. 6cm is close to 2 in, and a 2 in pipe is about \$17.75 per 10 feet (Home Depot (2025)). Since we require 52 m = 170 ft for the main tube, the piping for that section will only cost \$301. Given that the system uses the sun to heat up the water, the energy cost should be minimal, as the only energy needed would be for the devices, and they don't require a lot of energy.



(topographic-map.com, 2025)

This map shows the placement of each zeolite collector, as well as the placement of where the main pipe starts, with pipes coming from each collector to the main pipe.

If the zeolite machines are close together, they can block the air from reaching some of the machines, decreasing their efficiency. To solve this, they will be placed spread out, so that they don't interfere with the airflow to other machines. The zeolite devices will each be 16 m² or a 4 by 4 meter square. Since the machines will each be 16 m², 10 will be required to contain the necessary amount of zeolite to collect the necessary amount of water. Spreading out the machines poses a problem, however, as the main pipe can't go to all the machines at once, so to solve this, smaller pipes will be added between the main tube and each machine. According to the drawing, the machines are between 40 and 10 meters apart, with an average of around 21.5 meters away from the pipe. The total length of pipe required is 21.5*10=215 meters of piping, with a diameter of 6 cm. The 6 cm diameter is so that fewer diameters of pipe have to be

stored for leaks, decreasing the space needed to hold the repair materials. 215 meters is 706 feet, given the above cost, the new piping would cost \$1253.15. This added to the cost of the main pipe gives a total cost of piping of \$1554.15, which seems expensive except that the machines are almost energy-free, so the maintenance and run cost will be really small, justifying the large up-front cost. In addition, purchasing such a large amount of piping will most likely allow for a discount on the piping, decreasing the cost.

Water Storage & Distribution

Given that our sample population serviced by the project is 100 people and the average Algerian household has 4.7 people in it (*Algeria: average size of households 2010-2021*. (n.d.)), there are approximately 22 households in this region. Assuming that each household has one faucet dispensing at the United States' faucet flow rate upper limit (2.2 GPM), if every household ran their faucet at the same time, the flow rate would be 48.4 GPM. As it is unlikely every faucet in the region will ever be running at once, we can reduce this cap to about 70% of the maximum: 33.7 GPM.

Using a flow rate to pipe diameter chart (Roof, H.-T. (n.d.)), we can conclude that 35 GPM flowing via gravity at low pressure would require a pipe of a diameter 1 ½". This will be the diameter of pipe used to deliver water to the households.

According to the U.S. Department of Energy (*Best Management Practice #7: Faucets and Showerheads*. (n.d.)), federal standards mandate that all bathroom and kitchen faucets sold after January 1, 1994 must use no more than 2.2 gallons per minute. We will use that as our benchmark to estimate the maximum load for the household usage.

The average municipal water tower is sized to contain one day of the serviced region's water (*What Is The Purpose Of Water Towers?* (2022, December 16)). For this situation, we will accommodate one week of water to account for the inconsistency of the zeolite water generator and the fact that it can only produce water during daytime. According to Adam's calculations, the serviced region would consume approximately 640 Liters of water in one day. Multiplying this by 7 to find a week's consumption would equal 4480 Liters. Since one cubic meter of water equals 1000 Liters, we can approximate the size of the tank to be 4.5 m³. To minimize resources while building the tank, we will build it in the shape of a cylinder. Optimizing the surface area to a minimum value makes the height twice the size of the radius. This gives us a radius of 0.8947 meters, a height of 1.7894 meters, and a surface area of 15.0888 meters².

Using the surface area, we can calculate the material cost of the tank. We will assume it is a fully welded construction to negate the cost of bolts or rivets. According to a chart found online (*Galvanized Steel Water Tanks, Sectional Water Storage Tank, 4ft*4ft, Hot Dipped*. (n.d.)), the thickness of the steel used for a galvanized steel water tank is dependent on the tank's height. For a height of approximately 2 meters, each panel should be 3mm thick. According to an online listing (*Low Price Hot Dipped Galvanized Hot Cold Rolled Steel Plate Zinc Iron Sheet 3mm Metal*. (2024)), a 3mm thick roll of 12x1.5 meter (18m²) hot dipped galvanized hot/cold rolled steel costs \$610.00. Thanks to Alibaba's \$20 shipping cap on the first

order, the shipping cost is negligible. Assuming that the cost to cut, assemble, and transport this storage tank is an additional \$600, the cost to build this tank is \$1230.

Using google maps, the distance from the water tower to the center of the neighborhood is 860 meters. The distance from the center of the neighborhood to the midpoint between the center and edge of the neighborhood is ~250 meters. Multiplying this average distance from the center by the 22 households serviced by the zeolite machines gives us the total length of piping needed for the whole neighborhood: 5,500 meters. The total pipe distance is 6,360 meters. According to an Alibaba listing (*Agricultural Irrigation 1 / 1.5 / 1-1/4" / 2 / 3 Inch PVC Layflat Hose Flexible Water Pump Discharge Pipe*. (2021)), 6,360 meters of PVC pipe costs \$0.16 per meter, totaling \$1,017.16. However, the cheapest shipping option is \$1.38 per meter, totaling \$8735.55. The total cost of the piping would be \$9,793.96.

The water tower is placed ~5 meters lower than the elevation of the machines (1440-1450 meters), allowing gravity to carry the freshly generated water into the tank. We will also be adding a small filtration tank above the container, its volume will be ~30 liters or 0.03 cubic meters. Using the same formulas used for the main tank, the radius will be 0.1683 meters, the height will be 0.3366 meters, and the surface area will be 0.53391 m². There will be enough leftover galvanized steel from the primary tank to build this filtration tank.

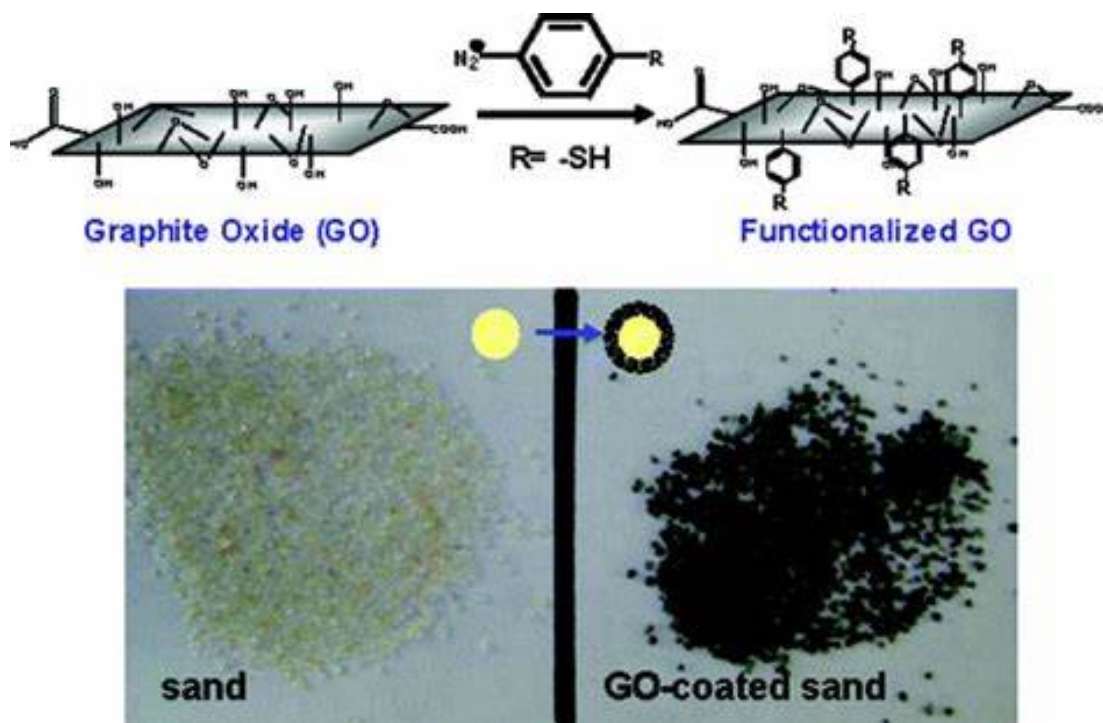
Filtration:

A major concern going into our design was creating an apparatus that allowed for effective and efficient filtration based on cleanliness and cost. One of the benefits of our proposed design is the simplicity it creates for filtration. Using zeolite allows for an efficient and natural filtration system which greatly reduces the cost of the project. This is an important component of the effectiveness of our design because water treatment is just as much of an issue in water starved areas as the accumulation of water itself. Atmospheric water harvesting is a relatively pure way of extracting water, especially using natural zeolite materials like we plan on doing. Zeolite also on its own is used in water filtration. Natural zeolites have a build up of negative charge around the molecule. This allows them to engage in cation exchange which takes potentially harmful heavy metals out of the water such as zinc and lead. In areas that struggle with water scarcity, heavy metals tend to be a major problem with the cleanliness of potential drinking water. Using zeolites is a cheap, natural solution to this problem. Zeolites can also be modified easily to allow for filtration of more harmful ions. The modified zeolites replace the outer negative charges with positive charges. This allows for an ion exchange of anions and adsorbs other organic materials.

Being able to harvest pure water allows for saving on filtration cost on the front end of water collection but still leaves room for improvement during transfer and storage of the water. The initial filtration from the zeolite does a great job removing the harmful ions and heavy metals but has no impact on protection from the debris that could be left from the transportation piping. In order to combat this, we propose a separate sand based filtration apparatus at the top of the water tower. The placement of the filter will use the otherwise designed infrastructure from the plan and will use no extra energy as the filter just runs based on gravity. Sand particles are

relatively small when compared to the contaminants that may be in the collected water and can pack really close together. When the sand particles are able to pack close together it leaves little room for anything other than pure water to pass through the gaps. Sand filters work by having an input valve and letting water run its course through the sand using gravity. The particles that are larger than the sand particles will be left in the filter while the pure water will be left on the other side and then ready to transport out.

Sand filters have been used around the world for a long time due to their efficiency. Sand filters are relatively simple to build and the materials for an effective filter can be found commonly. The filters also give compatibility to the design of the water collection apparatus as they can be engineered in many different sizes and complexities to account for the needs of the project. Additionally sand filters are comparatively inexpensive to other practical filtration methods. Sand filters however, aren't completely effective in removing all contaminants. That is why for our project we propose a variation of the traditional sand filter that has been engineered more recently, super sand filters. The name comes from the combining of two similar methods of filtration that have both proven to be effective, that is sand filtration and graphite filtration. Graphite particles leave even smaller gaps between themselves making for even more effective filtering. In super sand, engineers have developed sand particles that are coated in graphite oxide using nanotechnology. The graphite oxide coating around the sand allows the sand particles to pack even closer together making the GO-coated sand way more efficient at reducing the level of contaminants in the water.



(Nanowerk News.com. (2011))

Despite being more efficient at filtering water than normal sand, the filter rate is almost identical. In the lab, the GO-coated sand material was able to filter water contaminated with mercury from 400 parts per billion to under one in just over half an hour. The GO-coated sand is also relatively cheap compared to the amount it increases the efficiency of filtration. When bought by the ton, the modified sand is roughly double the cost of typical silica filtering sand. The dimensions of the filter that we will use in order to make sure that the filter will be able to keep up with the rates needed for the civilians we are serving are 42" x 96". This tank will more than suffice with a filtering capacity of 535 GPM flow rate. This rate is significantly more than needed for the current state of our proposal which gives the ability for expansion in the future, pending on the eventual success of our project. The proposed tank would be \$19,000. In order to fill the tank appropriately, it needs 3500 lbs of sand. Estimating the sand of Silica sand being \$13,000. Doubling this price for the sand to account for the cost increase of the modified sand, our estimated cost for the entire filter is \$45,000.

Benefits:

One of the major benefits of the zeolite water collection system is the improvement of health and safety in the community. The system will remove one reason to be outside in dangerous weather, will provide cleaner water as opposed to the groundwater currently used, and the water will automatically be filtered at no additional money or energy cost.

Algeria occasionally experiences dust storms, such as the one in 2022 (Guehaz, R., Sivakumar, V., & Mbatha, N. (2024)). These storms occur at least once a week during spring, when they are the most common (Goudie, A. S., & Middleton, N. J. (2001)). These dust storms are a major health hazard, as visibility is reduced, forcing people to slow down, the dust can irritate their lungs, and in worst-case scenarios, cause lung diseases. (Association, A. L. (n.d.)) Normally, people would have to enter the storms to get to the well so they could get their water, exposing them to this problematic weather phenomenon for a long time, as well as polluting the water further with all the particulate matter blown into it. However, if the zeolite water filtration system was implemented, the water would be pumped directly to the houses, meaning the people wouldn't have to venture outside to get water. This would keep them out of danger and keep them healthier, as they wouldn't be ingesting the dust. Also, since the zeolite systems close the openings in the side, not allowing air to enter during the day, the water inside wouldn't be affected by the dust storms during the day, keeping the water clean. Also, since the piping and tank are closed and the vulnerable piping is kept underground, the rest of the system would remain largely unaffected, ensuring clean water is delivered straight to houses in the community no matter the weather.

The community's health will also be improved due to the zeolite collection system because of how clean the collected water is. The current water used by the people in the affected area is groundwater, collected through wells. A study in 2024 studied drinking water in an area in Algeria with farming and living conditions similar to Abalessa, and the study found

that the water was, for the most part, unsafe to drink. Collecting data from wells, they determined that the water was only drinkable in 9.09% of the places tested. (Zair, N., Attoui, B., Miloudi, A., Abderrahmane, K., & Souyei, B. (2024)) This was calculated using the Synthetic Pollution Index (SPI), which is a method of finding how clean the water is based on the concentration of certain molecules in the water as compared to the WHO (World Health Organization) standard. Using the SPI, it was shown that 12 out of the 22 sites surveyed contained moderately polluted water (an SPI value of 0.5) or worse and only 2 of the 22 sites were deemed suitable for drinking. (Zair, N., Attoui, B., Miloudi, A., Abderrahmane, K., & Souyei, B. (2024)) For comparison, an SPI of 0.2 or less is required for the water to be suitable for drinking. According to this article, most of Algeria's groundwater is unsafe, and as such the zeolite water collection would benefit the health of the community, as they could drink clean water instead of trying to filter or boil the dirty water.

Water collected using the zeolite method, the water is automatically filtered. As stated in the approach section, the zeolite uses a cation exchange to remove heavy metals from the water. Also, as the water is collected from the air, it can't be contaminated by the ground like the groundwater can. The water gathered from the wells is very dirty, so collecting it from the air instead keeps the system from having to remove the pollutants found in the groundwater, decreasing the need for a filter. In addition, not having to remove pollutants in the water will most likely decrease the general stress level of the community, as they will no longer have to wonder if this water contains deadly bacteria or if they boiled it enough.

Overall, the installation of the zeolite water filtration system will increase not only the physical health and safety of the community but also improve the mental health of the community, helping them live better lives.

Another benefit of the zeolite water collection system is the increased economic opportunities for residents. Introducing new infrastructure to cities is a prime opportunity to introduce more jobs and economic opportunities to its residents. Building this zeolite water collection system will not only provide jobs upfront via skilled and unskilled labor in its installation, but will require maintenance from trained professionals, which could include graduates of the University of Tamanrasset, mere kilometers from the water tower's build site (to, T. (2019)).

The installation of such a water collection system would happen in stages, with surveyors working to ensure the ground on which the water tower and zeolite machines lie is level and stable. If this is not the case, laborers can come in to landscape the area. This work is ideal for young, unemployed men, a demographic that historically has had difficulty finding employment (6). In addition to leveling and stabilizing the area, ditches will have to be dug for the water pipes from the water generators to the water tower and from there to the spigots. This distance should only be ~500m but would be a sizable job due to the hill's size and incline. While it would be more economical to have the pipes run above ground, reducing upfront construction costs and improving ease of maintenance, the pipe chosen for this application is

PVC (Polyvinyl Chloride) is vulnerable to degradation and will become more brittle with regular ultraviolet exposure (7).

On the skilled workforce side of the project, local steelworkers and plumbers will be needed to create the ~4.5 m³ water storage tank and to design and fit the supply pipes feeding water from the zeolite generators to the storage tank and from the tank to the spigots. Local welding shops are common in Tamanrasset, and would provide business opportunities to local small businesses.

As for the ongoing jobs provided by this project, continual maintenance must be performed on the system, both the complex zeolite machines and the tower and pipes. This could be done initially as a co-op opportunity through universities like the University of Tamanrasset, which offer degrees in both management & STEM (8), making them ideal for producing future stewards of the water collection system. These graduates could leave school with full time job offers, monitoring and optimizing the production and throughput of the water to better provide for the residents who rely on this water.

Unintended consequences:

On a small scale, zeolite technology is extremely beneficial to solve local water insecurity issues, however scaling the technology may come with some issues. Zeolites are ores that are acquired through surface mining, usually by open pit mines (9). This form of mining can be dangerous and harmful to the environment. Open-pit mining is often used because of its relatively cheap costs. At large scales, the price for the equipment and running costs of open-pit mining is almost completely offset by the revenue generated from the ores. However, open-pit mining is extremely damaging to the environment. An example of this is the accumulation of waste. The extraction process leaves significant amounts of both solid and liquid waste which are just placed in disposal pits near the site. Another way in which open-pit mining can contaminate the surrounding environment is through dust cloud emissions. When the pit is being initially dug out, it is opened through explosives creating a crater-like hole. These explosives send huge clouds of debris into the air that sometimes contain harmful materials. Some of the dust that ends up in the air is from radioactive ores and other heavy metal materials. This debris has been linked to causing disease and other respiratory issues (10). If this contamination of the zeolite is significant it leaves back at square one of the problem; no better than the polluted groundwater. It is also possible however, that as mentioned before, the natural purifying benefit of the zeolite could make the contamination of the mining negligible.

The issues that mining for zeolite causes to the surrounding environment make it a hazard to any nearby community. Fortunately this would not be a problem for Abalessa, Algeria, since there are not any zeolite mines in the region. However, this brings a new difficulty into the equation. Another potential burden of using a zeolite water collection system is the cost of transportation of the zeolite. Zeolite mining is not known to be a significant part of industry in Algeria. In fact in the public mineral records there is no evidence of zeolite mining (10). Africa as

a whole is not a huge producer of zeolite minerals, which makes the transportation costs potentially significant.

10-Year Plan:

Over ten years, the zeolite water collection system in Abalessa will need to be maintained, but also should be expanded.

One major portion of the ten-year plan is the zeolite system at Abalessa will need to be maintained. For instance, the PVC piping used to transfer water from the zeolite collection machines to the tank needs to be replaced every 70 years or so. (4 Signs It's Time to Upgrade or Replace Your Plumbing. (2016, July 19)) (Kashmer, K. (2020, January 9)) This number is based on multiple sources, which all disagreed, so I took the average of their replacement times and it came out to about 70 years. The PVC will not need to be fully replaced during the ten-year plan, however, it will still need to be maintained. Rocks, dirt, and other things in the ground can scratch or puncture the PVC, meaning that sections may need to be replaced due to this damage. The main problem with this damage is, since the pipes are underground, the damage is all but invisible. To solve this, tests must be done of the water every year or so, checking for the heavy metals that contaminate the groundwater. For the tank holding the water, while it shouldn't have to be replaced over the course of the 10-year plan, it may get damaged due to dust storms, essentially sandblasting it or strong winds kicking up debris. As such, it must also be checked for damage periodically, to ensure the cleanliness of the drinking water for the families affected by the zeolite collection system. Thirdly, the zeolite machines must also be checked to make sure they are working. Since the zeolite collection system is a newer system, there is no data yet on how long they last, so maintenance checks should be made every month until a suitable baseline can be established. The final piece that would need replacement occasionally is the super sand filter. The filter should be replaced every 2-3 years, but may last longer if the filter shows no signs of damage and still filters water at about the same speed as at the start (E., M., & P.E., B. (2003, May)). The filter should be inspected weekly, but no major maintenance should have to be done besides replacement.

If the zeolite machinery performs well, then over ten years the system can be expanded. The top of the hill that would hold the zeolite is about 3200 m², so the zeolite, which only uses 160 m² per iteration, can easily be tripled or quadrupled in size without running into a space problem. The main issue with expanding the project is money. According to the approach section, the system costs about \$2.8 million per iteration, a massive cost. However, this money could be raised both by asking groups such as Oxfam, a global organization supporting many communities in all kinds of issues (Algeria | Oxfam International. (2022, May 25)). Also, after a year or so, hard data can be collected of exactly how well the system works, so people in the area and worldwide might be willing to help fund it since they know any money donated will go to a project that works.

A group is needed to oversee the project from Abalessa and maintain and expand it. One main place for candidates is the University of Tamanrasset, which is very close by. Some recent graduates could be hired to maintain and track the system, collecting data on how much water it actually generates and the water pressure, among other measurements. They could also have one or two people with business degrees working on procuring funding for more units and for the maintenance of the current ones. Maintenance on the pipes or other work requiring manual labor can be contracted out to residents in the area, allowing for the permanent team working the project to be small, cutting down on costs. A team stationed at the site is also helpful because they can work to improve it, maybe using smaller pipes or moving the zeolite machines around to find the best possible orientation and placement for each. This would increase the efficiency of the collection system, possibly reducing further costs as either fewer units or less piping are needed for the same amount of water to be collected.

Climate Change:

The transition to more sustainable water sources that aren't dominated by the effects of climate change is a great start for increasing the climate resilience of Abalessa, however as quality of living increases, energy consumption will also increase. This calls for significant exploration of clean energy. In 2015 energy consumption in Algeria was “about 58 MToe or 1.46 Toe / capita” and “the annual growth rate remained at about 7%” (Sahnoune, F., & Imessad, K. (2017)). This is especially detrimental to climate because Algeria, as discussed earlier, is a major exporter of natural gas and oil which results in “99% (of energy) sourced from natural gas”. This trends the country greatly towards intense air pollution. Due to the constant and intense sunlight in the region solar energy is strongly recommended and planned to be incorporated. As our efforts continue in these neighborhoods we intend to begin using solar to power various pumps and other incorporated infrastructure to keep our carbon output to a minimum. Algeria intends to increase its usage of renewable energy to “account for 27% of electrical generation” by 2030 (Sahnoune, F., & Imessad, K. (2017)). Algeria also intends to support field and forest rehabilitation which, while somewhat impractical in such a dry area, could eventually become a long term goal of the project, assuming water scarcity becomes less of a prominent issue due to the technology. This would add “a significant carbon sink and help to combat land degradation” throughout the entire area which could help to mitigate, and potentially reverse, drought effects and food scarcity in the region (Sahnoune, F., & Imessad, K. (2017)). In terms of individual acts our community can help accomplish, the largest would be help in the labor of installation of devices pertaining to these fields. Help in maintenance to the water collection devices, as discussed above, is a primary way to put the stepping stones of other improvements in place. Installation of solar panels on houses and in unused terrain would also be incredibly helpful to improve the lifestyle of the region. Finally, the inclusion and maintenance of plant life would ensure the slow recovery of the land in the fight against degradation ensued by climate change. There could also be great benefit from adapting “agricultural calendars to climate change” which can be done through the usage of drought-resistant crop varieties, crop diversification, changes in crop pattern and planting schedule” (Omran, E.-S. E. (2020)). While these techniques have been cited by many farmers

as ways to increase and improve yield in these intense conditions, there is little to no literature on how the “agricultural calendar is presently moving” (Omran, E.-S. E. (2020)). This is currently under research and is a massively important area to pay attention to in the coming years in order to improve food scarcity and refrain from the usage of fertilizers that pollute the natural watershed. Through all of these efforts the area can become more and more resilient to climate change as a whole and stand as a model for how to alleviate the climate challenges so many people face.

Conclusion:

Cost is a crucial component to the proposal of such a project. Both the up-front installation cost and the ongoing maintenance cost need to be taken into account. Fortunately, due to the solar and atmospheric nature of the zeolite water generators, there is no cost associated with the powering and operation of the machines. The only ongoing cost for the system is the continual upkeep and maintenance of the machines and water supply infrastructure. The zeolite’s cost, as described above, costs ~\$2.8 million, although there is room to lower this cost due to Algeria’s unique access to a special type of clay used to synthesize zeolite. The cost for 80m² of copper foil as a condensing surface costs ~\$7,600. The 1216ft² of wood used to frame and provide structure to the machines will cost ~\$3,600, bringing the total cost for the machines to \$2,811,200. This will be the majority of the cost for the project. The filtration tank will cost \$19,000 for the materials and assembly. Two options are available for the filter media; silica sand is commonly used in sand filters due to its granular structure and ability to filter out small contaminants. Approximately 3,500 pounds of sand would be required to fill the tank. Our other option for filter media is graphite oxide coated sand, also referred to as “Super Sand”. It filters contaminants at 2.5 times the rate of standard silica sand while only costing twice as much, bringing the top line cost for the filter to \$45,000 and the project’s total cost thus far to \$2,856,200. The cost of the water storage tank is \$630 for the materials and approximately another \$600 for the labor. The cost of the 6,360 meters of 1 ½” PVC piping from the tank to each of the 22 households supplied is \$1017.16, however the cheapest shipping option is direct from China at \$8,735.55. The cost of the 2” diameter PVC piping from the zeolite generators to the water tank is \$1,554.15, giving a total cost for the pipe at \$11,348.11 (Home Depot (2025)). This brings us to a net cost for the water collection and distribution system of \$2,969,778.11.

The proposed zeolite based solution to water collection provides solutions to many of the problems that face Abalessa, Algeria regarding water scarcity. Our solution blends together answers to environmental challenges in the area while effectively addressing the main concern of lack of clean water. Coming into this project we set out with the goal to not just come up with a generic solution, but to really immerse ourselves in the culture and issues of the area to come up with a design that would solve real life problems. Our design was made with the life of an everyday civilian in mind and what they would think is useful. This led to our design addressing the secondary issues of climate change and unemployment. On the water insecurity side, we focused on addressing the purity of the water and making sure the water would be pumped

directly into homes. With the implementation of our technology we hope the increased access to pure water will help the citizens of Abalessa, Algeria have more opportunities and be able to live a more fulfilled life.

Appendix of Calculations:

Energy requirement of the machine

$Q = mL_v$ (energy needed = mass of substance x latent heat of material)

(Latent Heat of Water: Definition, Formula, Forms of Latent Heat. (2025))

Primary tank

Volume:

$$V = \pi r^2 h$$

$$4.5 = \pi r^2 \cdot 2r$$

$$r = 0.8947 \text{ m}$$

Surface area:

$$A = 2\pi r h + 2\pi r^2$$

$$A = 2\pi \cdot 0.8947 \cdot 1.7894 + 2\pi \cdot 0.8947^2$$

$$A = 15.0888 \text{ m}^2$$

Miniature tank

Volume:

$$V = \pi r^2 h$$

$$0.03 = \pi r^2 \cdot 2r$$

$$r = 0.1683 \text{ m}$$

Surface area:

$$A = 2\pi r h + 2\pi r^2$$

$$A = 2\pi \cdot 0.1683 \cdot 0.3366 + 2\pi \cdot 0.1683^2$$

$$A = 0.53391 \text{ m}^2$$

Pipe from Machine to Storage Tank

Flow rate:

$$\frac{640L}{12h} = 53.3 \text{ L/h}$$

$$\frac{53.3}{60} = 0.88 \text{ L/min}$$

Energy loss due to friction:

50 m across, 16 m down, length = 52.49 m $\rightarrow \sqrt{50^2 + 16^2}$

Reynold's Number: $Dvp=0.06*3.847*10000.009204=267700$

Surface roughness = 0.0015 mm

Friction factor using moody diagram = 0.0144

Energy loss due to friction = $f * \frac{L}{D} * \frac{v^2}{2g} = (0.0144) * \frac{52.49}{3847} * \frac{.06^2}{2(9.8)} = 9.51m < 16m$

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